UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Applications of Linear Differential Equations: Simple Harmonic Oscillation

Complete study pack for class 015 (deck SESSION 11): the spring–mass model from Hooke's law and Newton's second law, $\frac{d^2x}{dt^2} + \omega^2 x = 0$ with $\omega = \sqrt{k/m}$, its solution $x = c_1 \cos \omega t + c_2 \sin \omega t = A \cos(\omega t + \varphi)$, amplitude, period, frequency and maximum velocity, the sign and unit conventions of spring problems, and the damped oscillator $\ddot{x} + 2\lambda n \dot{x} + n^2 x = 0$

Compiled from the PESU Unit 2 class 015 slide deck (23 slides), its AV-summary transcript (16.3 min), the Unit 1 & 2 class-work sheet (problem 37) and eight past ESA papers. Generated 17 October 2026.

· How to use this pack

Topic 1 builds the model: the forces on a hanging mass, the cancellation $mg = ks$ at equilibrium, the equation $\ddot{x} + \frac{k}{m}x = 0$ (E1.3) and the formulas for amplitude, period and frequency that the lecturer asks you to remember. Topic 2 solves spring problems: find m and k from the data, write $x(t)$, fix c_1, c_2 from the initial position and velocity with the sign convention (Figure F2.2), then report A, T, f, $v_{\max}$. It works both slide examples and CW P37. Topic 3 adds a resistive force: the damped oscillator of slides 015·15–16. The CF rules (complex roots) are in the class 002 pack and the LCR circuit, the electrical twin of this model, is in the class 014 pack (table C1.2). Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S015·11	Unit 2 deck 015 (“Applications of DE – SHM”), slide 11 (rendered .pptx)
T015 00:11:48	Unit 2 AV-summary transcript “Application problems on linear differential equations – Simple harmonic oscillation” (16.3 min), segment start
CW U2 P37	Unit 2 class-work problem 37

Contents

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5. Appendix B · Sources, coverage and gaps

1 The spring–mass model and simple harmonic motion

A mass m hangs from a spring of constant k. At equilibrium the weight balances the spring force, $mg = ks$. Displaced by x, the net force is $mg - k(s + x) = -kx$, so Newton's second law gives $\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$, a homogeneous LDE with constant coefficients. Its auxiliary roots are $\pm i\omega$ with $\omega = \sqrt{k/m}$, so $x = c_1 \cos \omega t + c_2 \sin \omega t = A \cos(\omega t + \varphi)$: simple harmonic motion with amplitude A, period $T = \frac{2\pi}{\omega}$ and frequency $\frac{1}{T}$.

Key terms and definitions

Equilibrium position. The position where the downward pull of gravity equals the upward restoring force of the spring: $mg = ks$. S015·5 · T015 00:02:29

Hooke's law. The restoring force is proportional to the displacement and acts in the opposite direction: $-k(s + x)$. S015·6 · T015 00:03:37

Spring (force) constant k. The proportionality constant in Hooke's law; force per unit stretch (lb/ft or N/m). S015·6 · T015 00:03:55

Simple harmonic motion. The motion $x(t) = c_1 \cos \omega t + c_2 \sin \omega t$ of the mass about its equilibrium position. S015·8 · T015 00:06:46

Angular frequency ω. $\omega = \sqrt{k/m}$, from writing the equation as $\frac{d^2x}{dt^2} + \omega^2 x = 0$. S015·7 · T015 00:05:33

Amplitude A. "A is called amplitude of the motion and gives the maximum displacement of the mass from its equilibrium position." S015·9 · T015 00:07:22

Period T. The time taken for one complete oscillation: $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}$. S015·9 · T015 00:07:40

Natural frequency f. The reciprocal of the period: $f = \frac{1}{T} = \frac{\omega}{2\pi}$ oscillations per second. S015·9 · T015 00:07:58

> *"From Hooke's Law, the restoring force is given by $-k(s + x)$ (the restoring force is proportional to displacement and acts in the opposite direction of displacement). By Newton's Second Law, the forces acting on the system $= ma$, i.e, $mg + (-k(s + x)) = ma$."* S015·6

> *"But at equilibrium, $mg = ks$. Therefore, $ma = -kx$. … $\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$. This equation can also be written as $\frac{d^2x}{dt^2} + \omega^2 x = 0$ where $\omega = \sqrt{\frac{k}{m}}$ called angular frequency."* S015·7

> *"The general solution of the differential equation is given by $x(t) = c_1 \cos \omega t + c_2 \sin \omega t$. $x(t)$ gives the position of the mass at any instant. The motion of the mass is Simple Harmonic Motion."* S015·8

Formulas

E1.1 · Equilibrium condition

ESA ★★

$$mg = ks$$

GIVES $k = \frac{mg}{s} = \frac{W}{s}$ from a static stretch s.

APPLIES The mass hangs at rest.

UNITS W in lb with s in ft (lb/ft); W in N with s in m (N/m).

NOTE $W = mg$ is the weight.

S015·5 · S015·7 · S015·11 · T015 00:02:29

E1.2 · Newton's second law for the hanging mass

ESA ★

$$mg - k(s + x) = m\frac{d^2x}{dt^2} \implies m\frac{d^2x}{dt^2} = -kx$$

GIVES The force balance; mg and ks cancel.

APPLIES x measured from equilibrium, downward positive.

S015·6 · S015·7 · T015 00:04:30 · T015 00:05:03

E1.3 · Equation of simple harmonic motion

ESA ★★

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0, \qquad \frac{d^2x}{dt^2} + \omega^2 x = 0, \quad \omega = \sqrt{\frac{k}{m}}$$

GIVES The model of the undamped, unforced spring.

APPLIES No damping, no external force.

NOTE A homogeneous LDE with constant coefficients: AE $m^2 + \omega^2 = 0$ (class 002, complex roots).

S015·7 · T015 00:05:15 · T015 00:05:33

E1.4 · General solution

ESA ★★

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

GIVES The position at any instant.

APPLIES Roots $\pm i\omega$.

NOTE $c_1 = x(0)$, $c_2 = \frac{\dot{x}(0)}{\omega}$.

S015·8 · T015 00:06:28 · T015 00:06:46

E1.5 · Amplitude–phase form

ESA ★★

$$x(t) = A\cos(\omega t + \varphi), \quad c_1 = A\cos\varphi,\ c_2 = -A\sin\varphi, \quad A = \sqrt{c_1^2 + c_2^2},\ \tan\varphi = -\frac{c_2}{c_1}$$

GIVES Amplitude and phase from c_1, c_2.

APPLIES After the constants are known.

NOTE Pick the quadrant of φ from the signs of $\cos\varphi = \frac{c_1}{A}$ and $\sin\varphi = -\frac{c_2}{A}$.

S015·8 · T015 00:07:05

E1.6 · Period and natural frequency

ESA ★★

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}, \qquad f = \frac{1}{T} = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

GIVES Seconds per oscillation; oscillations per second.

APPLIES Undamped motion.

NOTE The lecturer: “you should remember all these formulae”. The last form of f is added for clarity.

S015·9 · T015 00:07:40 · T015 00:07:58

E1.7 · Maximum velocity

ESA ★★

$$\dot{x} = -A\omega\sin(\omega t + \varphi), \qquad v_{\max} = A\omega$$

GIVES The greatest speed, reached at the equilibrium position.

APPLIES Any SHM.

NOTE Maximum acceleration is $A\omega^2$ (added for clarity).

S015·22 · CW U2 P37

Graphs and diagrams

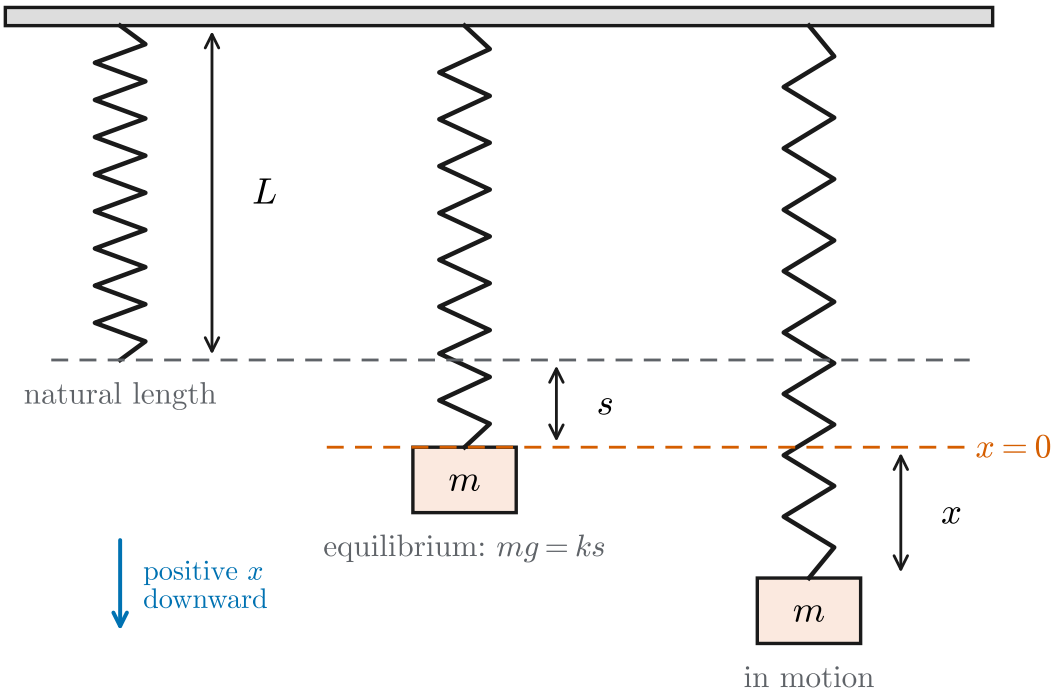
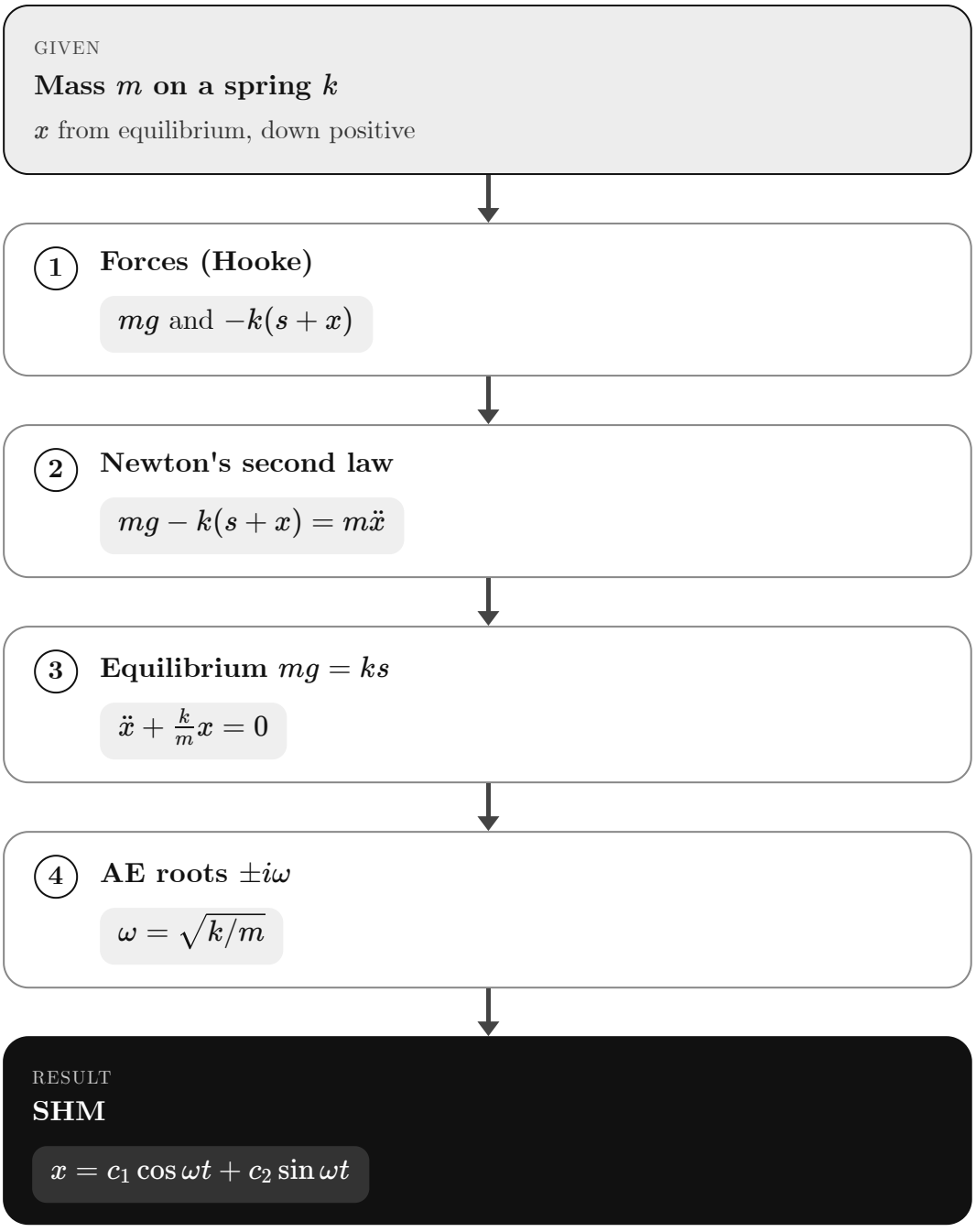


Figure G1.1. The spring in three positions (slide 015·5). ESA ★

S015·5 · T015 00:01:51

How to read it. Left: the natural length L. Middle: with the mass at rest the spring is stretched by s, where $mg = ks$; this is $x = 0$. Right: in motion the mass is x below equilibrium and the total stretch is $s + x$. Label L, s, x and $mg = ks$.

Flowcharts



Remark. Steps 4–5 are the complex-root case of class 002.

Figure F1.1. From forces to simple harmonic motion ESA ★

S015·6 · S015·7 · S015·8 · T015 00:04:30 · T015 00:05:33

The quantities of simple harmonic motion

Quantity	Formula	12 lb example	8 lb example
Angular frequency	$\omega = \sqrt{k/m}$	8 rad/s	$4\sqrt{3}$ rad/s
Amplitude	$A = \sqrt{c_1^2 + c_2^2}$	$\frac{5}{12}$ ft	$\frac{5}{12}$ ft
Period	$T = \frac{2\pi}{\omega}$	$\frac{\pi}{4}$ s	$\frac{\pi}{2\sqrt{3}}$ s
Frequency	$f = \frac{1}{T}$	$\frac{4}{\pi}$ Hz	$\frac{2\sqrt{3}}{\pi}$ Hz
Max velocity	$A\omega$	$\frac{10}{3}$ ft/s	$\frac{5}{\sqrt{3}}$ ft/s

S015·7 · S015·8 · S015·9 · S015·22

Spring–mass system and LCR circuit

Spring–mass	LCR circuit (class 014)
$m\ddot{x} + c\dot{x} + kx = F(t)$	$L\ddot{Q} + R\dot{Q} + \frac{Q}{C} = E(t)$
mass m	inductance L
damping coefficient c	resistance R
spring constant k	$\frac{1}{C}$
displacement x**, velocity** $\dot{x}$	charge Q, current I
$\omega = \sqrt{k/m}$	$\omega = \frac{1}{\sqrt{LC}}$

added for clarity S015·4 · T015 00:00:32

Important points and why

1. Two forces act on the mass: gravity mg (down) and the restoring force $-k(s+x)$; at equilibrium $mg = ks$, so only $-kx$ is left. ESA ★★ S015·6 · S015·7 · T015 00:03:37 · T015 00:05:03
2. x is measured from the equilibrium position, not from the natural length; that is why mg disappears from the equation. ESA ★★ S015·5 · T015 00:03:13
3. The equation is homogeneous with constant coefficients; its roots $\pm i\omega$ are purely imaginary, so the solution is the CF alone (no PI). ESA ★★ T015 00:05:53 · T015 00:06:10 · T015 00:06:28
4. "You should remember all these formulae": $\omega = \sqrt{k/m}$, $A = \sqrt{c_1^2 + c_2^2}$, $T = 2\pi\sqrt{m/k}$, $f = \frac{\omega}{2\pi}$. ESA ★★ S015·9 · T015 00:07:58
5. The period does not depend on the amplitude or on how the motion starts, only on m and k. ESA ★ added for clarity S015·9

Why does gravity not appear in $\ddot{x} + \frac{k}{m}x = 0$?
Because x is measured from equilibrium, where $mg = ks$ already. In $mg - k(s + x)$ the two constant forces cancel and only $-kx$ is left.

S015·7 · T015 00:05:03

Why are the motions sinusoidal?
The auxiliary equation $m^2 + \omega^2 = 0$ has roots $\pm i\omega$ with zero real part, so the CF is $c_1 \cos \omega t + c_2 \sin \omega t$: no growth and no decay.

T015 00:06:28 · S015·8

Exam trap. Amplitude is $\sqrt{c_1^2 + c_2^2}$ (a sum of squares), not "c_1 square by c_2 square" as said at T015 00:07:22. ADDED FOR CLARITY S015·8 · T015 00:07:22

Exam trap. At equilibrium $mg = ks$ with the static stretch s, not kx (T015 00:02:46): x is the displacement from equilibrium. ADDED FOR CLARITY S015·5 · T015 00:02:46

Worked examples

W1.1 · Deriving the equation of motion (slides 015·5–9) ESA ★

GIVEN A mass m hangs from a spring of constant k; s is the stretch at equilibrium and $x(t)$ the displacement from equilibrium (down positive).

STEP 1 Forces: gravity mg and the restoring force $-k(s + x)$ (Hooke's law). Newton: $mg - k(s + x) = m\ddot{x}$.

STEP 2 At equilibrium $mg = ks$, so $m\ddot{x} = -kx$, i.e. $\ddot{x} + \frac{k}{m}x = 0 = \ddot{x} + \omega^2 x$ with $\omega = \sqrt{k/m}$.

STEP 3 AE $r^2 + \omega^2 = 0$, roots $\pm i\omega$: $x = c_1 \cos \omega t + c_2 \sin \omega t$.

STEP 4 Put $c_1 = A \cos \varphi$, $c_2 = -A \sin \varphi$: $x = A \cos(\omega t + \varphi)$, period $\frac{2\pi}{\omega}$.

ANSWER $\boxed{\frac{d^2x}{dt^2} + \omega^2 x = 0,\ x = A\cos(\omega t + \varphi),\ T = 2\pi\sqrt{\frac{m}{k}}}$

CHECK sympy dsolve of $\ddot{x} + \omega^2 x = 0$ returns $c_1 \cos \omega t + c_2 \sin \omega t$.

S015·5 · S015·6 · S015·7 · S015·8 · S015·9 · T015 00:03:37 · T015 00:05:33 · T015 00:07:05

SHORT-NOTE RECAP

Spring–mass: $mg = ks$ at equilibrium; Hooke $-k(s + x)$; Newton $m\ddot{x} = -kx$, so $\ddot{x} + \omega^2 x = 0$, $\omega = \sqrt{k/m}$. Roots $\pm i\omega$: $x = c_1 \cos \omega t + c_2 \sin \omega t = A\cos(\omega t + \varphi)$, $A = \sqrt{c_1^2 + c_2^2}$, $\tan \varphi = -\frac{c_2}{c_1}$. $T = \frac{2\pi}{\omega} = 2\pi\sqrt{m/k}$, $f = \frac{1}{T}$, $v_{\max} = A\omega$.

2 Solving spring problems

Every spring problem follows one recipe: get m from the weight ($m = W/g$) and k from the static stretch ($k = W/s$), write $\ddot{x} + \omega^2 x = 0$, fix c_1 from the initial position and c_2 from the initial velocity, then report the amplitude, period, frequency and maximum velocity $A\omega$. Units must agree (feet with $g = 32$ ft/s², metres with $g = 9.8$ m/s²), and positive x is downward. Class 015 works a 12 lb and an 8 lb weight; the class-work sheet adds a 10 kg body in SI units.

Key terms and definitions

Free motion. Motion with no external force and no damping: $\ddot{x} + \frac{k}{m}x = 0$. S015·10 · T015 00:08:48

Formulas

E2.1 · Mass from weight ESA ★★

$$m = \frac{W}{g}, \qquad g = 32 \text{ ft/s}^2 \text{ or } 9.8 \text{ m/s}^2$$

gives	Mass in slugs (from lb) or kg.
applies	The weight is given in pounds or kilogram-weight.
note	A 12 lb weight has $m = \frac{12}{32} = \frac{3}{8}$ slug; a body of 10 kg has $m = 10$ kg.

S015·11 · S015·18 · T015 00:10:05

E2.2 · Constants from the initial conditions ESA ★★

$$c_1 = x(0), \qquad c_2 = \frac{\dot{x}(0)}{\omega}$$

gives	The particular motion.
applies	Downward positive: pulled down d → $x(0) = +d$; raised d → $x(0) = -d$; upward velocity v → $\dot{x}(0) = -v$; released → $\dot{x}(0) = 0$.

S015·13 · S015·14 · S015·20 · T015 00:14:07 · T015 00:14:24

Graphs and diagrams

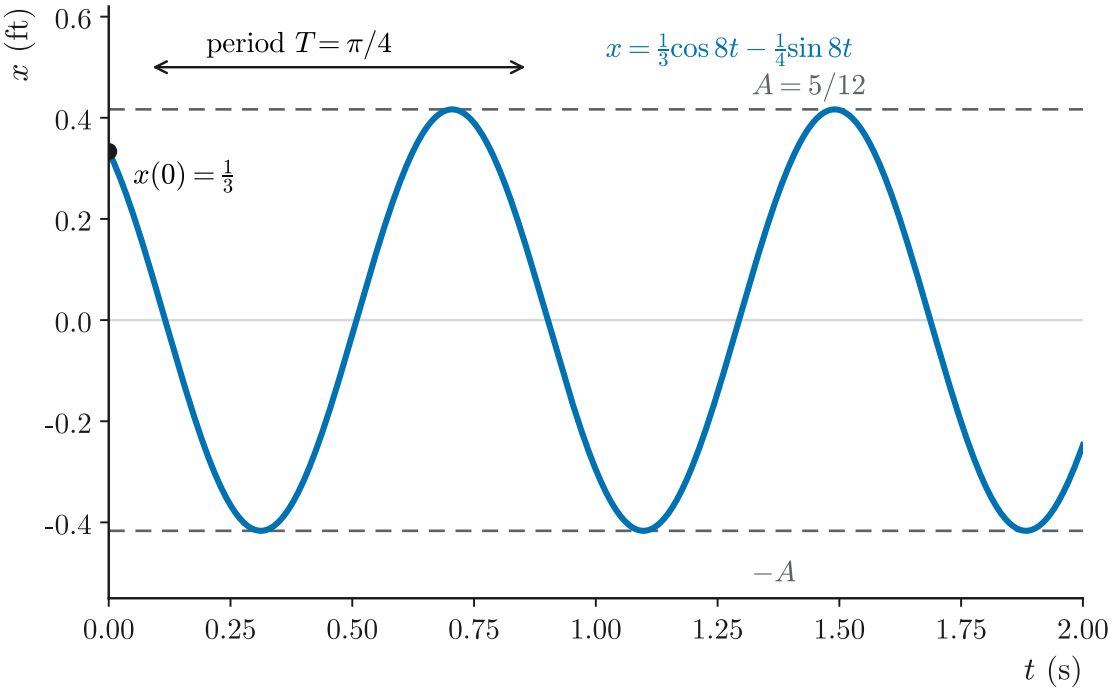


Figure G2.1. Motion of the 12 lb weight: $x = \frac{1}{3}\cos 8t - \frac{1}{4}\sin 8t$. ESA ★★

ADDED FOR CLARITY S015·14

How to read it. The mass starts $\frac{1}{3}$ ft below equilibrium (black dot) moving upward, so the curve starts downward-sloping on the plot. It swings between $\pm\frac{5}{12}$ ft (grey dashed) with period $\frac{\pi}{4} \approx 0.785$ s; the first crossing of 0 is at 0.116 s (W2.2). Positive x is below equilibrium.

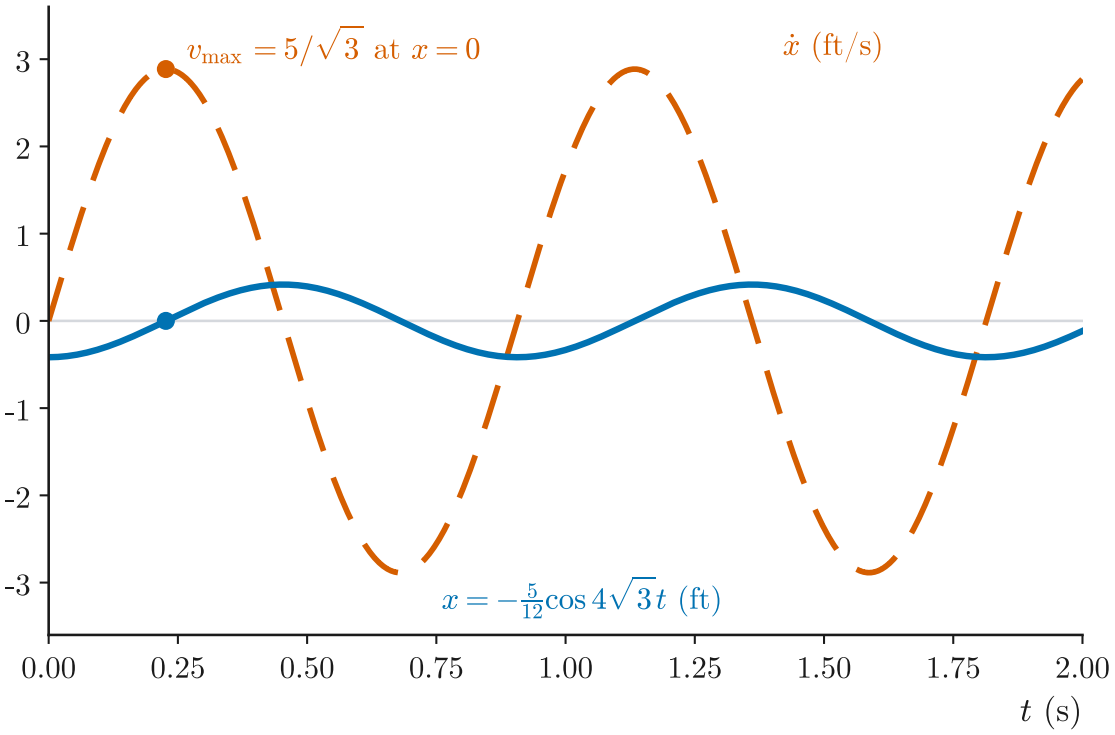
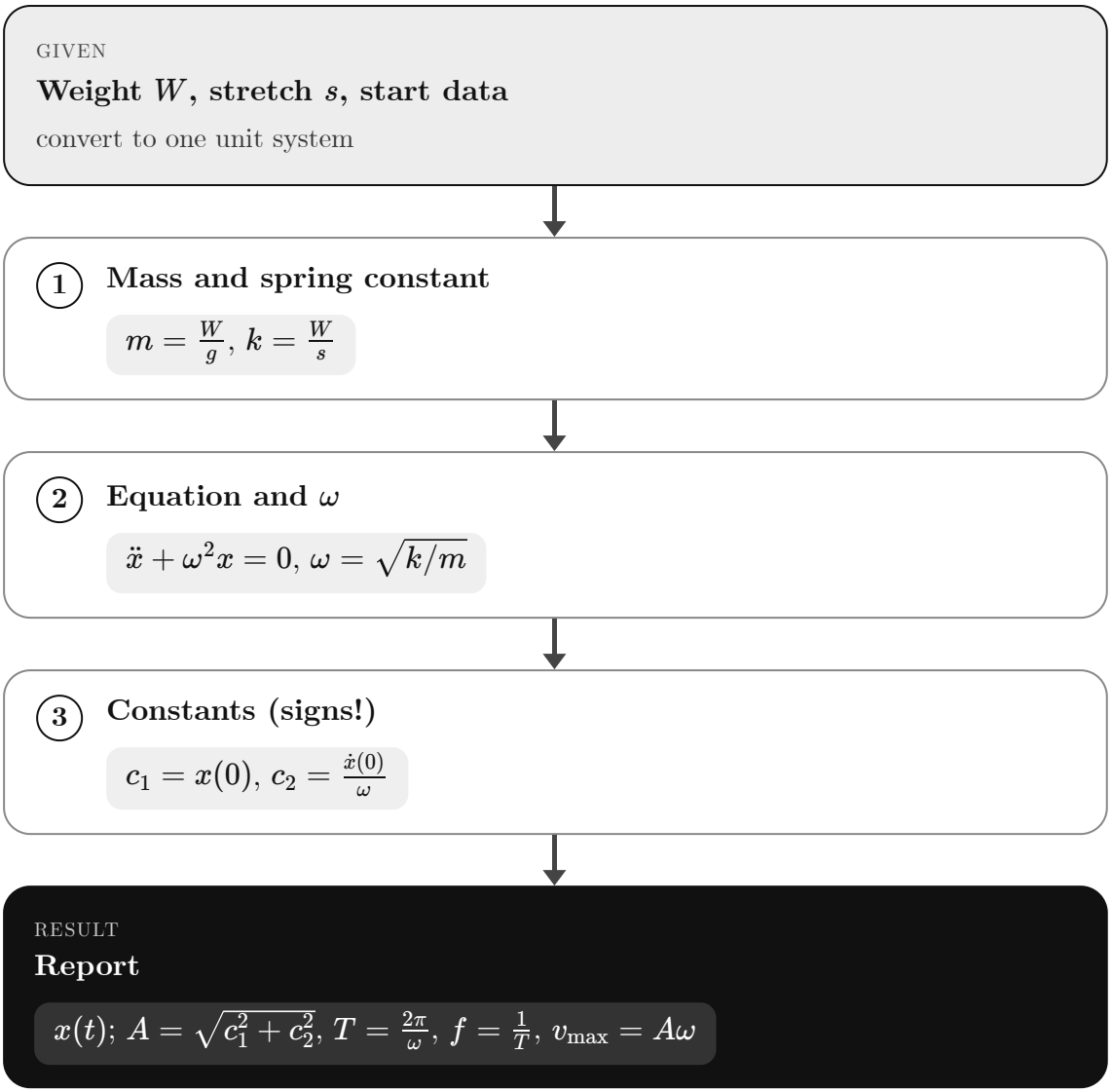


Figure G2.2. Displacement and velocity of the 8 lb weight. ESA ★★

ADDED FOR CLARITY S015·21 · S015·22

How to read it. Blue: $x = -\frac{5}{12}\cos 4\sqrt{3}t$ (ft). Vermilion dashed: $\dot{x} = \frac{5\sqrt{3}}{3}\sin 4\sqrt{3}t$ (ft/s). The velocity is largest, $\frac{5}{\sqrt{3}} \approx 2.89$ ft/s, exactly where $x = 0$, and zero at the turning points $x = \pm\frac{5}{12}$.

Flowcharts

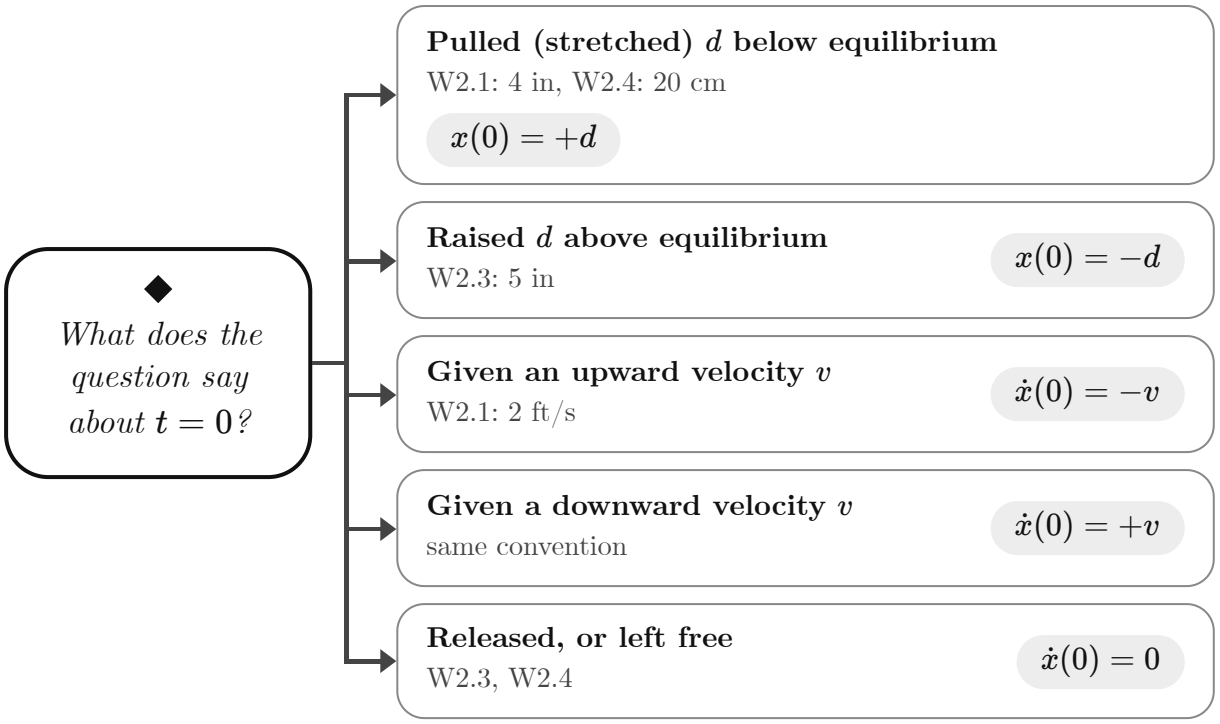


Remark. W2.1, W2.3 and W2.4 all follow these four steps.

Exam trap. Inches to feet before using $g = 32$ ft/s².

Figure F2.1. Solving a spring problem ESA ★★

S015·10 · S015·11 · S015·12 · S015·13 · S015·14 · T015 00:09:25 · T015 00:12:41



Remark. The convention is stated in the lecture and on slide 015·20.

Figure F2.2. Turning the words into initial conditions (down positive) ESA ★★

S015·13 · S015·14 · S015·20 · T015 00:14:07 · T015 00:14:24

Every spring problem of this class, side by side

Problem	m, k	Equation	$x(0)$, $\dot{x}(0)$	$x(t)$
12 lb, 6 in (W2.1)	$\frac{3}{8}$, 24	$\ddot{x} + 64x = 0$	$\frac{1}{3}$, -2	$\frac{1}{3}\cos 8t - \frac{1}{4}\sin 8t$
8 lb, $k = 12$ (W2.3)	$\frac{1}{4}$, 12	$\ddot{x} + 48x = 0$	$-\frac{5}{12}$, 0	$-\frac{5}{12}\cos 4\sqrt{3}t$
CW P37 (W2.4)	10, 1960	$\ddot{x} + 196x = 0$	0.2, 0	$0.2\cos 14t$

S015·10 · S015·17 · CW U2 P37

Important points and why

1. Use the static data first: the weight gives $m = W/g$, and the stretch it causes gives $k = W/s$. The later data (pulled down, velocity) fix c_1, c_2. ESA ★★ S015·11 · T015 00:09:25
2. Convert inches to feet (12 in = 1 ft) before using $g = 32$ ft/s²: 6 in = 0.5 ft, 4 in = $\frac{1}{3}$ ft, 5 in = $\frac{5}{12}$ ft. ESA ★★ S015·11 · S015·13 · S015·20 · T015 00:10:47
3. Downward is positive. An upward velocity of 2 ft/s is $\dot{x}(0) = -2$; a mass raised 5 in has $x(0) = -\frac{5}{12}$ ft. ESA ★★ S015·14 · S015·20 · T015 00:14:07 · T015 00:14:24
4. Differentiate $x(t)$ before using a velocity condition: $\dot{x} = -c_1\omega\sin\omega t + c_2\omega\cos\omega t$, so $\dot{x}(0) = c_2\omega$. ESA ★★ S015·13 · T015 00:13:31

5. Released from rest means $\dot{x}(0) = 0$, so $c_2 = 0$ and $x = x(0)\cos\omega t$; the amplitude is then $|x(0)|$. ESA ★★ S015·20 · S015·21 · CW U2 P37
6. The maximum velocity $A\omega$ is reached when the mass passes through equilibrium. ESA ★★ ADDED FOR CLARITY S015·22 · CW U2 P37

Why is an upward velocity negative?
The deck takes downward displacement as positive; a velocity in the opposite (upward) direction therefore carries a minus sign.
T015 00:14:07 · T015 00:14:24 · S015·20

Exam trap. Mixing inches and feet: with $g = 32$ ft/s², every length must be in feet ($s = 6$ in $= 0.5$ ft gives $k = 24$; using 6 gives $k = 2$, wrong). ADDED FOR CLARITY S015·11

Exam trap. $c_2 = \frac{\dot{x}(0)}{\omega}$, not $\dot{x}(0)$: in the 12 lb example $8c_2 = -2$ gives $c_2 = -\frac{1}{4}$ (the lecture first says "minus 4", T015 00:14:59). ADDED FOR CLARITY S015·14 · T015 00:14:59

Exam trap. Weight is not mass: a 12 lb weight has $m = \frac{12}{32}$ slug. In SI, a 20 kg weight is a force of $20g = 196$ N. ADDED FOR CLARITY S015·11 · CW U2 P37

Exam trap. A body raised above equilibrium starts at a negative x ($-\frac{5}{12}$ ft on slide 015·20); forgetting the sign flips the phase. ADDED FOR CLARITY S015·20

Worked examples

W2.1 · 12 lb weight, 6 in stretch, pulled 4 in down, upward 2 ft/s (slides 015·10–14) ESA ★★

GIVEN A spring is stretched 6 in by a 12 lb weight. The weight is pulled down 4 in below equilibrium and given an upward velocity of 2 ft/s. No damping. Find the motion, the amplitude and the period.

STEP 1 $m = \frac{12}{32} = \frac{3}{8}$ slug; $s = 6$ in $= 0.5$ ft, $k = \frac{mg}{s} = \frac{12}{0.5} = 24$ lb/ft.

STEP 2 $\ddot{x} + \frac{24}{3/8}x = \ddot{x} + 64x = 0$, $\omega = 8$: $x = c_1\cos 8t + c_2\sin 8t$.

STEP 3 $x(0) = 4$ in $= \frac{1}{3}$ ft gives $c_1 = \frac{1}{3}$.

STEP 4 $\dot{x} = -\frac{8}{3}\sin 8t + 8c_2\cos 8t$; upward velocity: $\dot{x}(0) = -2$, so $8c_2 = -2$, $c_2 = -\frac{1}{4}$.

STEP 5 $A = \sqrt{\frac{1}{9} + \frac{1}{16}} = \frac{5}{12}$ ft; $T = \frac{2\pi}{8} = \frac{\pi}{4}$ s.

ANSWER $\boxed{x(t) = \frac{1}{3}\cos 8t - \frac{1}{4}\sin 8t \text{ ft};\ A = \frac{5}{12} \text{ ft};\ T = \frac{\pi}{4} \text{ s}}$

CHECK sympy dsolve with both initial conditions agrees; the slides agree.

S015·10 · S015·11 · S015·12 · S015·13 · S015·14 · T015 00:08:33 · T015 00:11:48 · T015 00:15:34

W2.2 · Amplitude–phase form of W2.1, and when the mass first passes equilibrium

ESA ★

GIVEN Write $x = \frac{1}{3}\cos 8t - \frac{1}{4}\sin 8t$ as $A\cos(8t+\varphi)$ and find the first $t > 0$ with $x = 0$.

STEP 1 $A = \frac{5}{12}$; $\cos\varphi = \frac{c_1}{A} = \frac{4}{5}$, $\sin\varphi = -\frac{c_2}{A} = \frac{3}{5}$, so $\varphi = \tan^{-1}\frac{3}{4} = 0.6435$ rad.

STEP 2 $x = \frac{5}{12}\cos(8t + 0.6435)$; first zero when $8t + 0.6435 = \frac{\pi}{2}$.

ANSWER $\boxed{x = \frac{5}{12}\cos(8t+0.6435);\ \text{first crossing at } t = \dfrac{\pi/2 - 0.6435}{8} \approx 0.116 \text{ s}}$

CHECK nsolve of $\frac{1}{3}\cos 8t - \frac{1}{4}\sin 8t = 0$ near 0.1 gives 0.11591.

ADDED FOR CLARITY S015·8 · S015·14

W2.3 · 8 lb weight, $k = 12$ lb/ft, raised 5 in and released (slides 015·17–22)

ESA ★★

GIVEN An 8 lb weight hangs from a spring with $k = 12$ lb/ft. It is raised 5 in above equilibrium and released. Find the equation of motion, $x(t)$, the amplitude, period, frequency and maximum velocity.

STEP 1 $m = \frac{8}{32} = \frac{1}{4}$ slug; $\frac{1}{4}\ddot{x} + 12x = 0$, i.e. $\ddot{x} + 48x = 0$.

STEP 2 $r^2 + 48 = 0$, $r = \pm 4\sqrt{3}\,i$, $\omega = 4\sqrt{3}$: $x = C_1\cos 4\sqrt{3}t + C_2\sin 4\sqrt{3}t$.

STEP 3 Raised 5 in: $x(0) = -\frac{5}{12}$ ft gives $C_1 = -\frac{5}{12}$; released: $\dot{x}(0) = 4\sqrt{3}C_2 = 0$, $C_2 = 0$.

STEP 4 $A = \frac{5}{12}$ ft, $T = \frac{2\pi}{4\sqrt{3}} = \frac{\pi}{2\sqrt{3}}$ s, $f = \frac{2\sqrt{3}}{\pi}$ Hz, $v_{\max} = A\omega = \frac{5}{12}\cdot 4\sqrt{3} = \frac{5}{\sqrt{3}}$ ft/s.

ANSWER $\boxed{x(t) = -\frac{5}{12}\cos 4\sqrt{3}\,t = \frac{5}{12}\sin\left(4\sqrt{3}\,t - \frac{\pi}{2}\right);\ A = \frac{5}{12},\ T = \frac{\pi}{2\sqrt{3}} \approx 0.907 \text{ s},\ f \approx 1.10 \text{ Hz},\ v_{\max} = \frac{5}{\sqrt{3}} \approx 2.89 \text{ ft/s}}$

CHECK sympy dsolve agrees; the two forms on slide 015·21 are equal.

S015·17 · S015·18 · S015·19 · S015·20 · S015·21 · S015·22

W2.4 · 10 kg body, 20 kg weight stretches 10 cm, pulled 20 cm down (CW P37) ESA ★★

GIVEN A body weighing 10 kg hangs from a spring. A pull of 20 kg weight stretches the spring 10 cm. The body is pulled 20 cm below equilibrium and released. Find $x(t)$, the maximum velocity and the period.

STEP 1 $k = \frac{20g}{0.1} = 200g = 1960$ N/m (taking $g = 9.8$ m/s²); $m = 10$ kg.

STEP 2 $\omega^2 = \frac{k}{m} = 196$, $\omega = 14$ rad/s: $\ddot{x} + 196x = 0$, $x = c_1 \cos 14t + c_2 \sin 14t$.

STEP 3 $x(0) = 0.2$ m gives $c_1 = 0.2$; released: $\dot{x}(0) = 14c_2 = 0$, $c_2 = 0$.

STEP 4 $v_{\max} = A\omega = 0.2 \times 14 = 2.8$ m/s; $T = \frac{2\pi}{14} = \frac{\pi}{7}$ s.

ANSWER $\boxed{x(t) = 0.2\cos 14t \text{ m};\ v_{\max} = 2.8 \text{ m/s};\ T = \frac{\pi}{7} \approx 0.449 \text{ s}}$

CHECK sympy dsolve agrees. The CW sheet prints no answer. Check: $\omega^2 = \frac{k}{m} = \frac{20g}{0.1\times 10} = 20g = 196$.

CW U2 P37

SHORT-NOTE RECAP

Recipe: $m = W/g$, $k = W/s$ (feet with $g = 32$, metres with $g = 9.8$); $c_1 = x(0)$, $c_2 = \dot{x}(0)/\omega$ with down positive. 12 lb, 6 in, 4 in down, 2 ft/s up: $x = \frac{1}{3}\cos 8t - \frac{1}{4}\sin 8t$, $A = \frac{5}{12}$ ft, $T = \frac{\pi}{4}$ s. 8 lb, $k = 12$, raised 5 in: $x = -\frac{5}{12}\cos 4\sqrt{3}t$, $v_{\max} = \frac{5}{\sqrt{3}}$. CW P37: $x = 0.2\cos 14t$, $v_{\max} = 2.8$ m/s, $T = \frac{\pi}{7}$.

3 The damped oscillator

A resistance proportional to velocity, $2\lambda mn\dot{x}$, adds a first-derivative term: $\ddot{x} + 2\lambda n\dot{x} + n^2x = 0$. For $0 < \lambda < 1$ the roots are $-\lambda n \pm in\sqrt{1-\lambda^2}$ and the motion is an oscillation whose amplitude decays like $e^{-\lambda nt}$. Slides 015·15–16 solve it from $x(0) = 0$, $\dot{x}(0) = x_0$. It is the mechanical twin of the LCR circuit of class 014.

Key terms and definitions

Damping (resistive) force. A force opposing the motion and proportional to the velocity; on slide 015·15 its magnitude is $2\lambda mn\dot{x}$. S015·15

Formulas

E3.1 · Damped oscillator ESA ★

$$m\ddot{x} + 2\lambda mn\dot{x} + mn^2x = 0 \implies \ddot{x} + 2\lambda n\dot{x} + n^2x = 0$$

GIVES The model with a resistance $2\lambda mn\dot{x}$.
APPLIES Restoring force mn^2x (so $n = \sqrt{k/m}$).
S015·15

E3.2 · Roots and damped frequency ESA ★

$$r = -\lambda n \pm in\sqrt{1-\lambda^2}, \qquad \omega_d = n\sqrt{1-\lambda^2}$$

GIVES The underdamped CF $e^{-\lambda nt}(A\cos\omega_d t + B\sin\omega_d t)$.
APPLIES $0 < \lambda < 1$.
NOTE Slide 015·16 calls ω_d simply ω.
S015·16

E3.3 · Damped motion from $x(0) = 0$, $\dot{x}(0) = x_0$ ESA ★

$$x(t) = \frac{x_0}{n\sqrt{1-\lambda^2}}\, e^{-\lambda nt} \sin\left(n\sqrt{1-\lambda^2}\, t\right)$$

GIVES The slide's boxed answer.
APPLIES $0 < \lambda < 1$.
NOTE Checked by substitution and both initial conditions (sympy).
S015·16

Graphs and diagrams

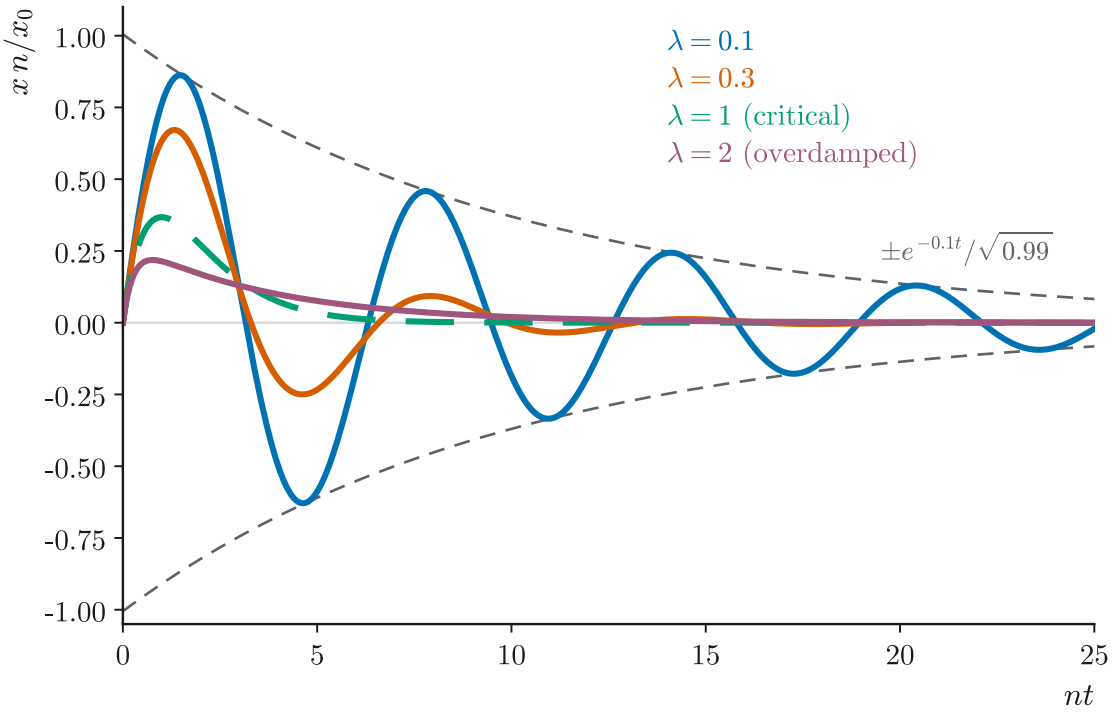
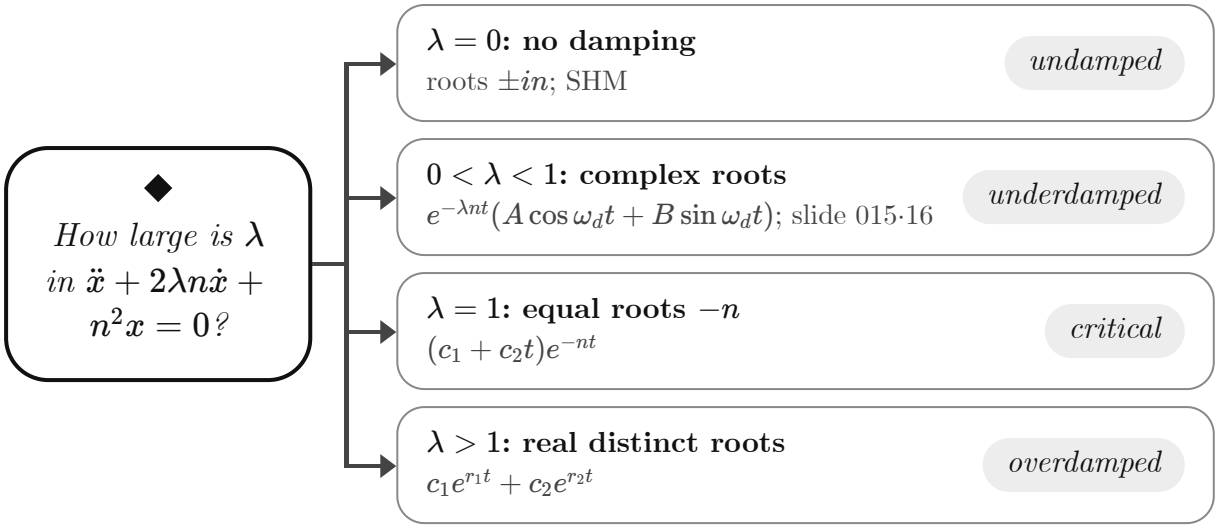


Figure G3.1. Damped motion for $\lambda = 0.1$, 0.3, 1 and 2 ($n = 1$, $x_0 = 1$). ESA ★

ADDED FOR CLARITY S015·16

How to read it. Underdamped curves ($\lambda = 0.1$ blue, 0.3 vermilion) oscillate inside their envelopes $\pm\frac{e^{-\lambda t}}{\sqrt{1-\lambda^2}}$ (grey dashed for $\lambda = 0.1$). Critical ($\lambda = 1$, green) and overdamped ($\lambda = 2$, purple) rise once and creep back without crossing zero.

Flowcharts



Remark. Only the first two oscillate.

Figure F3.1. Damped oscillator: which motion? ESA ★

ADDED FOR CLARITY S015·15 · S015·16

Undamped and damped oscillators

	Undamped	**Damped, $0 < \lambda < 1$**
Equation	$\ddot{x} + n^2 x = 0$	$\ddot{x} + 2\lambda n\dot{x} + n^2 x = 0$
Roots	$\pm in$	$-\lambda n \pm in\sqrt{1-\lambda^2}$
Frequency	n	$n\sqrt{1-\lambda^2}$ (smaller)
Amplitude	constant A	decays like $e^{-\lambda nt}$
$x(0) = 0,\ \dot{x}(0) = x_0$	$\frac{x_0}{n}\sin nt$	$\frac{x_0}{n\sqrt{1-\lambda^2}} e^{-\lambda nt}\sin(n\sqrt{1-\lambda^2}t)$

S015·7 · S015·15 · S015·16

Important points and why

1. A resistance proportional to velocity adds the term $2\lambda n\dot{x}$; for $0 < \lambda < 1$ the mass still oscillates, but inside the envelope $\pm Ce^{-\lambda nt}$. ESA ★ S015·15 · S015·16
2. Damping lowers the frequency: $\omega_d = n\sqrt{1-\lambda^2} < n$. ESA ★ ADDED FOR CLARITY S015·16
3. With $\lambda = 1$ (critical) or $\lambda > 1$ (overdamped) the roots are real and the mass returns to equilibrium without oscillating. ESA ★ ADDED FOR CLARITY S015·16

Why is the damped motion called a twin of the LCR circuit?
$m\ddot{x}+c\dot{x}+kx=F(t)$ and $L\ddot{Q}+R\dot{Q}+\frac{Q}{C}=E(t)$ are the same equation with $m\leftrightarrow L$, $c\leftrightarrow R$, $k\leftrightarrow\frac{1}{C}$ (table C1.2).

ADDED FOR CLARITY S015·4 · T015 00:00:32

Worked examples

W3.1 · Damped oscillator with $x(0)=0$, $\dot{x}(0)=x_0$ (slides 015·15–16) ESA ★

GIVEN A particle of mass m moves on a line under a force mn^2x towards a fixed point O, with resistance $2\lambda mn\dot{x}$, $0<\lambda<1$. Initially $x(0)=0$, $\dot{x}(0)=x_0$. Find $x(t)$.

STEP 1 Newton: $m\ddot{x}+2\lambda mn\dot{x}+mn^2x=0$, i.e. $\ddot{x}+2\lambda n\dot{x}+n^2x=0$.

STEP 2 $r^2+2\lambda nr+n^2=0$: $r=-\lambda n\pm in\sqrt{1-\lambda^2}$; $x=e^{-\lambda nt}(A\cos\omega t+B\sin\omega t)$, $\omega=n\sqrt{1-\lambda^2}$.

STEP 3 $x(0)=0$ gives $A=0$; $\dot{x}(0)=B\omega=x_0$ gives $B=\frac{x_0}{n\sqrt{1-\lambda^2}}$.

ANSWER
$$\boxed{x(t)=\frac{x_0}{n\sqrt{1-\lambda^2}}\,e^{-\lambda nt}\sin\left(n\sqrt{1-\lambda^2}\,t\right)}$$

CHECK Substitution leaves 0; $x(0)=0$ and $\dot{x}(0)=x_0$ (sympy). Matches slide 015·16.

S015·15 · S015·16

W3.2 · The same problem with $\lambda=1$ and $\lambda=2$ ESA ★

GIVEN Solve $\ddot{x}+2\lambda n\dot{x}+n^2x=0$, $x(0)=0$, $\dot{x}(0)=x_0$ for $\lambda=1$ and for $\lambda>1$.

STEP 1 $\lambda=1$: double root $-n$; $x=(c_1+c_2t)e^{-nt}$, $c_1=0$, $c_2=x_0$.

STEP 2 $\lambda>1$: roots $-\lambda n\pm n\sqrt{\lambda^2-1}$ (real); $x=\frac{x_0}{2n\sqrt{\lambda^2-1}}\left(e^{r_1t}-e^{r_2t}\right)$.

ANSWER
$$\boxed{\lambda=1:\ x=x_0te^{-nt};\quad \lambda>1:\ x=\frac{x_0}{n\sqrt{\lambda^2-1}}\,e^{-\lambda nt}\sinh\left(n\sqrt{\lambda^2-1}\,t\right)}$$

CHECK Both satisfy the equation and the initial conditions (sympy); neither crosses $x=0$ again (Figure G3.1).

ADDED FOR CLARITY S015·16

SHORT-NOTE RECAP

Damped: $\ddot{x}+2\lambda n\dot{x}+n^2x=0$; $0<\lambda<1$: roots $-\lambda n\pm in\sqrt{1-\lambda^2}$, decaying oscillation; from $x(0)=0$, $\dot{x}(0)=x_0$: $x=\frac{x_0}{n\sqrt{1-\lambda^2}}e^{-\lambda nt}\sin(n\sqrt{1-\lambda^2}t)$. $\lambda\geq1$: no oscillation.

A Symbol glossary

Symbol	Name	Meaning	Unit	Appears in
m	Mass	W/g; slug (ft–lb) or kg. Not the AE variable here (the AE uses r)	slug, kg	E1.3, E2.1
W	Weight	mg; pounds or kilogram-weight	lb, N	E2.1
g	Acceleration due to gravity	32 ft/s² or 9.8 m/s²	ft/s², m/s²	E1.1, E2.1
k	Spring constant	Force per unit stretch	lb/ft, N/m	E1.1, E1.3
s	Static stretch	Extension at equilibrium, $mg = ks$	ft, m	E1.1
$x(t)$	Displacement	From equilibrium, downward positive	ft, m	E1.3, E1.4
$\dot{x}$, $\ddot{x}$	Velocity, acceleration	$\frac{dx}{dt}$, $\frac{d^2x}{dt^2}$		E1.2, E2.2
ω	Angular frequency	$\sqrt{k/m}$; on slide 015·16 the damped frequency	rad/s	E1.3, E3.2
A	Amplitude	$\sqrt{c_1^2 + c_2^2}$; on slide 015·16 an integration constant	ft, m	E1.5
φ	Phase	$\tan\varphi = -\frac{c_2}{c_1}$	rad	E1.5
T, f	Period, frequency	$\frac{2\pi}{\omega}$, $\frac{1}{T}$	s, Hz	E1.6
c_1, c_2	Arbitrary constants	$x(0)$, $\dot{x}(0)/\omega$		E1.4, E2.2
n	Undamped frequency	In mn^2x (slide 015·15)	rad/s	E3.1
λ	Damping ratio	$0 < \lambda < 1$ on the slide		E3.1, E3.2
x_0	Initial velocity	$\dot{x}(0)$ on slide 015·15 (not a position)	ft/s	E3.3

B Sources, coverage and gaps

Source	Path	Use
Deck 015 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/015_Applications_of_DE-_SHM__07c768aa6b33cecd.*	23 slides; content on 4–22 (every equation read from the rendered images)
Transcript T015 (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/015_Application_problems_on_Linear_differential_equations_-_Simple_harmonic_oscillation__9f24b70546186809.*	16.3 min; slides 4–14
Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Unit 2 problem 37 (no printed answer; solved here)
ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	Checked: no SHM question in any paper (shared ESA map)
Video 015 (.mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/015_Application_problems_on_Linear_differential_equations_-_Simple_harmonic_oscillation__9f24b70546186809.mp4	Not opened: every slide equation is readable and no video trigger applies

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024 (19 distinct Unit 2 questions, 24 occurrences). Simple harmonic oscillation (the spring–mass problem) was never asked in any of them: 0 of 8 papers. It is in the syllabus ("Application problems – ... Oscillations of a spring") and the class-work sheet has one problem (CW P37), so the method is ★★ and the formulas $\omega = \sqrt{k/m}$, $T = 2\pi/\omega$, $f = 1/T$ are ★★ because the lecturer stresses them ("you should remember all these formulae", T015 00:07:58). The derivation and the damped oscillator are ★. The applications question the papers do ask is the LCR circuit (class 014, Jan 2024 sets 3 and 4).

Known gaps, conflicts and corrections

1. Every slide answer in this class is right. Slides 015·10–14 ($m = \frac{3}{8}$, $k = 24$, $\omega = 8$, $x = \frac{1}{3}\cos 8t - \frac{1}{4}\sin 8t$, $A = \frac{5}{12}$ ft, $T = \frac{\pi}{4}$ s), slides 015·15–16 (damped, $x = \frac{x_0}{n\sqrt{1-\lambda^2}} e^{-\lambda n t}\sin(n\sqrt{1-\lambda^2}\,t)$) and slides 015·17–22 ($x = -\frac{5}{12}\cos 4\sqrt{3}t$, $T = \frac{\pi}{2\sqrt{3}}$, $f = \frac{2\sqrt{3}}{\pi}$, $v_{\max} = \frac{5}{\sqrt{3}}$) were recomputed with sympy dsolve and the initial conditions.
2. Slide 015·11 gives $g = 32$ "ft/sec"; the unit of an acceleration is ft/s² (slide 015·18 has it right).
3. Slides 015·4 and the lecture (T015 00:00:32) say a mass attached to a vertical "string"; it is a spring.
4. Slide 015·6 spells Hooke's law "Hook's Law", and so does the lecture.
5. The lecture reads the amplitude as "square root of c_1 square by c_2 square" (T015 00:07:22); slide 015·8 has $A = \sqrt{c_1^2 + c_2^2}$, a sum.
6. The lecture says "C_2 is equal to minus 4" (T015 00:14:59) before writing $x = \frac{1}{3}\cos 8t - \frac{1}{4}\sin 8t$; $c_2 = -\frac{1}{4}$, as on slide 015·14.

7. The lecture says "Mg is equal to Kx at equilibrium position" (T015 00:02:46); it is $mg = ks$ (the static stretch s, not the displacement x), as on slides 015·5 and 015·7. It also calls ks "the displacement of the spring"; s is the displacement and ks the spring force.
8. The lecture's natural-frequency sentence is cut off ("so 1 by 2", T015 00:08:15); slide 015·9 gives $\frac{1}{T} = \frac{\omega}{2\pi}$, which with $\omega = \sqrt{k/m}$ is $\frac{1}{2\pi}\sqrt{\frac{k}{m}}$ (added for clarity).
9. Slides 015·15–22 have no lecture. The damped oscillator (slides 15–16) is solved on the slides only for $0 < \lambda < 1$; the critical ($\lambda = 1$) and overdamped ($\lambda > 1$) cases in topic 3 are added for clarity.
10. Slide 015·21 writes the particular solution as $\frac{5}{12}\sin(4\sqrt{3}t - \frac{\pi}{2})$; this equals $-\frac{5}{12}\cos 4\sqrt{3}t$ (checked).
11. The unit sign convention (downward positive, so "pulled down" gives $x(0) > 0$ and an upward velocity gives $\dot{x}(0) < 0$) is stated in the lecture (T015 00:14:07–00:14:42) and on slide 015·20, not on slides 10–14.
12. CW P37 (10 kg body, 20 kg weight stretches the spring 10 cm, pulled 20 cm down and released) has no printed answer on the sheet. Solved here with $g = 9.8$ m/s² and kilogram-weight forces: $k = 1960$ N/m, $\omega = 14$ rad/s, $x = 0.2\cos 14t$ m, $v_{\max} = 2.8$ m/s, $T = \frac{\pi}{7} \approx 0.449$ s. The sheet says "stretch the spring to 10 cm", read as "by 10 cm".
13. No SHM question appears in the eight ESA papers checked.
14. Transcript recognition errors were corrected silently: "XFT"/"XFP" → $x(t)$, "Hook's law" → Hooke's law, "case" → ks, "regulatory motion" → oscillatory motion, "seconds law" → second law, "Consider see a big question" → consider an example, "feed" → feet, "Kn" → k and m, "Pass of the object" → mass of the object, "C even cos A T" and "sine A T" → $c_1\cos 8t$, $\sin 8t$, "plus r minus i omega" → $\pm i\omega$, "one by 3 cos C8" → $\frac{1}{3}\cos 8t$.

Coverage

Source	Citations	Slides cited (slide × citations)
S015	109	4×2, 5×6, 6×7, 7×11, 8×10, 9×8, 10×4, 11×8, 12×2, 13×6, 14×8, 15×6, 16×10, 17×2, 18×2, 19×1, 20×8, 21×3, 22×5
CW	6	
T015	56	

Slides 1–3 and 23 are title, session, class-content and closing slides. The transcript (T015, 16.3 min) teaches slides 4–14 and ends with the 12 lb example; slides 15–22 (the damped oscillator and the 8 lb example, typeset differently) have no lecture and are cited by slide alone. Bucket numbers are not contiguous: the deck's title slide says SESSION 11, and the bucket numbers 007, 008, 010, 012 and 013 do not exist (009 = session 7, 011 = sessions 8 and 9, 014 = 10, 015 = 11).